

Deployment Readiness Memo

System Summary

This project developed a facial verification system using a ResNet18-based embedding model trained with Triplet Loss. The system was audited for demographic fairness across gender, age, and skin tone groups.

The audit measured False Accept Rate (FAR) and False Reject Rate (FRR) for both overall performance and demographic subgroups.

Audit Findings

The initial audit revealed performance inconsistencies across demographic groups. Several subgroups showed elevated FAR values, indicating uneven verification reliability.

The fairness audit pipeline successfully identified subgroup-level disparities and enabled targeted fairness analysis.

Bias Mitigation

A balanced sampling mitigation strategy was implemented to reduce demographic imbalance during training.

After mitigation:

- subgroup metrics became more differentiated
- overall accuracy slightly improved
- fairness evaluation became more stable

Performance Trade-Offs

Initial Model Accuracy

0.5000

Mitigated Model Accuracy

0.5042

The mitigation strategy slightly improved fairness-aware performance while preserving overall system functionality.

Remaining Ethical Risks

Despite mitigation efforts, important risks remain:

- High FAR values for certain demographic groups
- Limited dataset diversity
- Potential demographic underrepresentation
- Risk of misuse in surveillance or law-enforcement contexts

The system should therefore be considered experimental and not production-ready for high-stakes applications.

Final Recommendation

This system demonstrates a complete responsible AI auditing workflow and fairness evaluation pipeline.

However, due to remaining fairness disparities and moderate verification performance, deployment in high-stakes environments is not currently recommended without substantial additional validation and model improvements.